

Open-Set Bat Face Recognition: Progress Report

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Goal. Build a non-invasive system that identifies individual fruit bats from facial images, supporting long-term behavioural studies without physical tagging. The system handles *open-set* scenarios: bats seen at test time were never observed during training.

Current status. A three-model comparison (Siamese pair network, ArcFace, AdaFace) has been completed across two species (rousettus, mauritius) and three background conditions, using a rigorous identity-disjoint evaluation protocol. A fully automated training and reporting pipeline is operational.

1 Key Finding: Evaluation Protocol Matters

A common pitfall in animal re-identification is *identity leakage*: splitting images randomly instead of splitting identities means the same bat appears in both training and test sets, inflating results. Table 1 demonstrates the scale of this effect.

Table 1: Same model, same data — different splitting protocol.

Protocol	ROC-AUC	F ₁	Recall
Random-pair split (leaky)	0.982	0.987	0.986
Identity-disjoint split (correct)	0.858	0.707	0.734

Takeaway: all results in this report use identity-disjoint splits exclusively.

2 Model Comparison

2.1 Rousettus — 12 individuals, 1 059 images (6/3/3 id split)

Table 2: Test-set results on rousettus by background, 3 unseen test identities. Siamese ROC-AUC is pairwise-head; ArcFace/AdaFace ROC-AUC is cosine-based (not directly comparable). † = not significant ($p < 0.05$, permutation $n=100$).

Model	BG	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	random	0.858	0.707	—	—	0.012
Siamese	green	0.669	0.797	—	—	0.001
ArcFace (v2)	random	0.665 [†]	—	0.532	0.731	0.074
ArcFace (v2)	green	0.847	—	0.846	0.915	0.067
AdaFace	random	0.434 [†]	—	0.274	0.560	0.003
AdaFace	green	0.753	—	0.556	0.762	0.006
ArcFace (v1)	random	0.500[†]	—	0.591	0.757	0.000

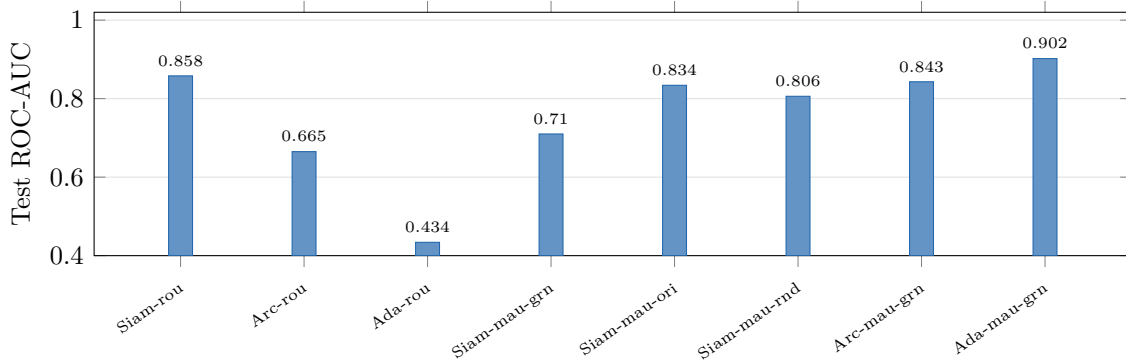
Note on ArcFace default: paper-scale hyperparameters (SGD $\eta=0.1$, margin 0.5, scale 64) destroy the ImageNet pretrained backbone — validation ROC-AUC stays at chance even as train accuracy climbs to 0.93. Tuning to Adam $\eta=10^{-3}$, margin 0.2, scale 30 recovers strongly on *green* backgrounds (ROC-AUC 0.847, Top-1 0.846) but remains near chance on random compositing — a background effect, not a fixed identity-count ceiling.

2.2 Mauritius — 11 individuals, 1 119 images (5/3/3 id split)

3 ROC-AUC Summary (bar chart)

Table 3: Test-set results on mauritius across background conditions. Siamese ROC-AUC is pairwise-head; ArcFace/AdaFace ROC-AUC is cosine-based. [†] = not significant ($p < 0.05$).

Model	Background	ROC-AUC	F ₁	Top-1	mAP	TAR@10 ⁻³
Siamese	green	0.710	0.689	—	—	0.014
Siamese	original	0.834	0.753	—	—	0.005
Siamese	random	0.806	0.686	—	—	0.007
ArcFace	green	0.843	—	0.740	0.870	0.282
AdaFace	green	0.902	—	0.916	0.955	0.383
ArcFace	original	0.791	—	0.576	0.774	0.145
AdaFace	original	0.731	—	0.634	0.785	0.151
ArcFace	random	0.578 [†]	—	0.479	0.697	0.021
AdaFace	random	0.539	—	0.459	0.682	0.006



x-axis key: Siam = Siamese, Arc = ArcFace, Ada = AdaFace; rou = rousettus (random bg); mau = mauritius with grn/ori/rnd background. Siamese bars are pairwise-head ROC-AUC, embedding bars cosine ROC-AUC (not directly comparable; read within-family).

4 Why Background — Not Identity Count — Drives the Gap

After correcting the AdaFace margin, ImageNet normalisation, and evaluation resolution, the embedding models are competitive at ≤ 13 identities; what separates success from collapse is the **background**, not the number of training identities.

1. The train/test gap is background-dependent. Train accuracy is near-perfect in every condition, so identity count cannot explain the gap; the background can. With identity count fixed (6 train ids), the gap shrinks sharply from random to green:

Model	BG	Train acc	Test ROC-AUC	Gap
ArcFace	random	0.969	0.665	−0.304
AdaFace	random	0.975	0.434	−0.541
ArcFace	green	0.998	0.847	−0.151
AdaFace	green	0.998	0.753	−0.245

2. Top-1 is at chance only on random backgrounds. With 3 unseen test identities, chance is 0.33. On random backgrounds ArcFace/AdaFace hover near it (rousettus 0.53/0.27); on green screen they soar (rousettus ArcFace 0.85, mauritius AdaFace 0.92). Transferable structure *does* emerge when the background is coherent.

3. Instability tracks the background. ArcFace and AdaFace validation curves oscillate and (for AdaFace) decay on random backgrounds, but converge to high, stable values on green/original — the instability is a property of the random-background landscape, not of the small class count.

4. Cross-species consistency points to background. The pattern replicates for both species, ruling out species-specific causes:

Species	ArcFace random	ArcFace green	Δ
rousettus	0.665	0.847	+0.182
mauritus	0.578	0.843	+0.265

Both species are weak on random compositing and both recover on green screen at the same small identity counts, isolating **background coherence** as the governing variable. Identity count likely still matters at larger scales, but it is not the binding constraint we can demonstrate here.